

Challenge 10: The Unfriendly Partition Conjecture

Definition 1 (Partial partition). A *partial partition* of the vertices of a graph G is a partial function $V(G) \rightarrow \{0, 1\}$. We call it *finite* if its domain is finite. We call it a *partition* if it is a total function.

Definition 2 (Unfriendly). Let v be a vertex of a graph G . A partial partition $f: V(G) \rightarrow \{0, 1\}$ is *unfriendly at v* if v has at least as many neighbours mapped by f to the opposite partition class as it has neighbours mapped to the same partition class. Formally this means

$$|N(v) \cap f^{-1}[1 - f(v)]| \geq |N(v) \cap f^{-1}[f(v)]|.$$

A partition is *unfriendly* if it is unfriendly at every vertex.

Conjecture 1.1 (Unfriendly partition Conjecture). Every countable graph has an unfriendly partition.

In order to pose a scaled version of this conjecture, we introduce the following game:

Definition 3 (The r -partitioning game.). Let r be an ordinal number and let G be a graph. The *r -partitioning game on G* is played between two players, Challenger and Partitioner, as follows. A position in the game is a pair (f, α) consisting of a partial partition f of G and an ordinal number α . The initial position is $(\emptyset, r)$. In each turn, beginning in position (f, α) , Challenger must name a vertex v of G and an ordinal $\beta < \alpha$ and Partitioner must then name a finite partial partition g extending f which is defined at v and unfriendly at v . After this, the new position is (g, β) . If at any point some player is unable to move, the other player wins.

Note that since there are no infinite decreasing sequences of ordinals this game must end after finitely many turns.

Conjecture 1.2 (Scaled Unfriendly Partition Conjecture). Let r be an ordinal. For any countable graph G , Partitioner has a winning strategy in the r -partitioning game on G .